



## Aim

Recognise scientific processes of fermentation and simple control factors.

## INTRODUCTION

Winemaking is based on one simple biochemical reaction that converts sugar to alcohol. When a sugar (such as fructose or glucose) is combined with yeast bacteria and allowed to ferment, it changes form to produce ethyl alcohol plus carbon dioxide and energy.

Components of grape juice that are important for wine production

- Sugars – glucose, fructose
- Organic acids – tartaric acid, malic acid, citric acid
- Tannins – catechol, chlorogenic acid, caffeic acid
- Amino acids and proteins
- Minerals – phosphates, sulphates
- Vitamins – B-group, ascorbic acid
- Important volatile components that affect flavour and colour

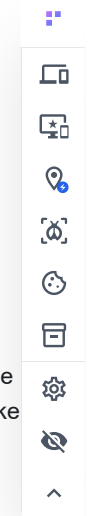
## FERMENTATION

Fermentation is very important in winemaking. It is what makes a wine and distinguishes it from grape juice. Wild yeasts start the fermentation process, but they only produce about 4% alcohol and impart a yeasty, bitter taste to the wine. Wine yeasts then take over, changing the remaining sugar into alcohol (bringing the alcohol content to about 10%). If left unchecked, vinegar bacteria then take over, turning the wine into vinegar.

The winemaker must remove unwanted wild yeasts and vinegar bacteria if he or she is to control the fermentation process and produce quality wine. Potassium bisulphite is added to the must to kill unwanted microorganisms and prevent oxidation.

The sugars that provide the basic substrate for the reaction to occur are found in the grapes. Even though yeast is found naturally on the surface of grapes, “wine yeast” is added to the crushed grape pulp to produce the type of fermentation required for quality wine production. The important characteristic of yeasts is metabolising sugars to form alcohol without using oxygen. This process is also referred to as anaerobic respiration. Certain commonly found bacteria are also capable of anaerobic respiration but with the undesirable result of acetic acid (vinegar) or other souring compounds. If present, these bacteria will lead to unpleasant odours and tastes in the wine, hence the importance of clean, uncontaminated equipment.

The speed of fermentation is dependent on temperature. In many commercial wineries, the temperature of the fermentation tanks is carefully controlled to produce steady temperatures for optimum fermentation. Yeasts only survive temperatures between 4°C and 32°C. The closer to 4°C, the slower they work; the closer to 32°C, the faster they work. White wines are generally fermented in temperatures in the range of 12° to 18°C; reds are fermented at higher temperatures. Nevertheless, it is important to keep the



temperature of the fermenting must from rising too high. Cooler conditions are preferable. 'Off' flavours may be produced when the fermentation reaction raises the temperature of large quantities of must to levels which are too high. If you make smaller quantities of wine and the temperature is kept cooler such flavour problems are less likely.

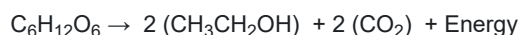
A short fermentation period produces a light, inexpensive wine. A longer fermentation time produces a finer quality wine.

During the process of fermentation, sugar in the form of glucose (C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>) is converted to alcohol (CH<sub>3</sub>CH<sub>2</sub>OH) and carbon dioxide gas (CO<sub>2</sub>). Fermentation is induced by yeast. You will note from the chemical formula that glucose contains six carbon, six oxygen and twelve hydrogen atoms.

Although this reaction seems straight forward, the reactions which take place inside a yeast cell which enable fermentation to occur are complex.

## Fermentation as a Biochemical Reaction

The overall process can be summarised below:



|         |           |                  |
|---------|-----------|------------------|
| Glucose | Alcohol   |                  |
| (sugar) | (ethanol) | (carbon dioxide) |

The conversion of glucose to alcohol requires the action of 12 enzymes. Ten enzymes are required to convert glucose to pyruvic acid, and a further two enzymes known as pyruvate decarboxylase and alcohol dehydrogenase are needed to convert pyruvic acid into ethanol (also known as ethyl alcohol) and carbon dioxide.

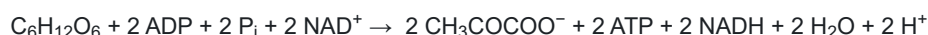
The first 10 steps in alcoholic fermentation is the same process as glycolysis in the human body (which produces lactic acid from glucose during prolonged exercise).

The difference is that whilst both processes involve the same 10 enzymes for the first 10 steps to produce pyruvate (pyruvic acid), there is only one further step in glycolysis, and this involves the enzyme lactate dehydrogenase which converts pyruvate to lactic acid.

In alcoholic fermentation there are two different enzymes involved which convert pyruvate into ethanol and carbon dioxide i.e. pyruvate decarboxylase and alcohol dehydrogenase.

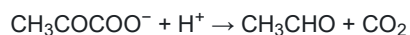
## Glycolysis

The formula for glycolysis is as follows:

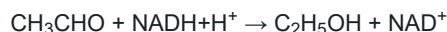


In the above formula pyruvate is denoted by CH<sub>3</sub>COCOO<sup>-</sup>, and inorganic phosphate is denoted by Pi.

In the final two steps pyruvate is converted to ethanol and carbon dioxide. Pyruvate is first catalysed by pyruvate decarboxylase:



Secondly, it is catalysed by alcohol dehydrogenase:



It should be noted that oxygen is not needed during fermentation. In fact, its presence will cause some yeasts to oxidise pyruvate into carbon dioxide and water, but not alcohol. This is because these yeasts can only produce alcohol in anaerobic environments. In the presence of oxygen cellular respiration causes oxidation of pyruvate. Nevertheless, there are some yeast types that will produce alcohol from pyruvate in the presence of oxygen, such as baker's yeast (*Saccharomyces cerevisiae*).

## CARBOHYDRATES

The process of fermentation begins with sugar molecules which are carbohydrates. Carbohydrates are compounds that contain carbon combined with hydrogen and oxygen. They are one of the most significant groups of organic compounds that are made (i.e. synthesised) by living organisms and are very important to them.

Carbohydrates are compounds which all have empirical formulae which are multiples of the simple formula  $\text{CH}_2\text{O}$ . Chemically, carbohydrates are defined as "polyhydroxy aldehydes" or "ketones". Examples of carbohydrates include sugars, starch, glycogen, cellulose and chitin.

There are three main types of carbohydrates: monosaccharides, oligosaccharides, and polysaccharides.

### a) Monosaccharides

These are simple sugars which under reasonably mild conditions can be hydrolysed into smaller compounds. There are four common monosaccharides which include glucose and fructose which are typically used in fermentation:

Glucose ( $\text{C}_6\text{H}_{12}\text{O}_6$ ) - a building block for many other compounds, and one of the most common compounds on earth.

Fructose - ( $\text{C}_6\text{H}_{12}\text{O}_{12}$ ) - contains six carbon, six oxygen and twelve hydrogen atoms the same as glucose, but they are arranged or bonded together in a different pattern with a five member ring as opposed to a six member ring found in sucrose. It is found together with glucose in honey and fruit juices.

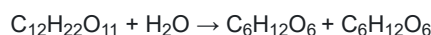
Ribose - occurs naturally as part of vitamin B12 and the ribonucleic acids (RNA).

2-Deoxyribose - occurs naturally as part of DNA (i.e. deoxyribonucleic acid) which is an essential part of the genetic material found in all living organisms.

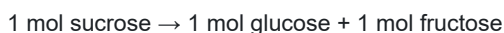
### b) Oligosaccharides

These are compound sugars which when hydrolysed (broken down due to reaction with water) will yield two to six molecules.

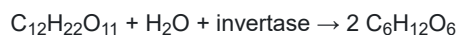
Disaccharides - (double sugars) are oligosaccharides which yield two monosaccharide molecules through hydrolysis or due to the action of specific enzymes in the digestive tract:



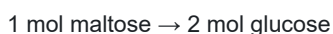
Sucrose - is what we usually know as table sugar. It is extracted as ordinary sugar cane or beet sugar and is about twice as sweet as glucose. Rum is made from the fermentation and distillation of sugar cane. On hydrolysis, sucrose forms glucose and fructose:



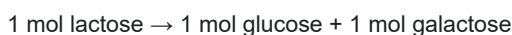
In the first step of alcoholic fermentation, the enzyme known as invertase is used to break down the glycosidic linkage between glucose and fructose molecules:



Maltose - is formed by the breakdown of starch during digestion and the manufacture of malt liquors. On hydrolysis:



Lactose - occurs in milk and is the only one of the common sugars not found in plants. It is less soluble and less sweet than other sugars. On hydrolysis:



### c) Polysaccharides

These are compounds that yield many monosaccharides when they hydrolyse. Polysaccharides include starches, cellulose and related materials. These consist of many monosaccharide units which may be alike or different.

Starch - is insoluble and a stable and efficient store of potential energy in plants where it is found in roots, seeds and stems. It is a mixture of microscopic granules of two polysaccharides - amylose and amylopectin. These contain many glucose units (amylose 70-350 and amylopectin 20-30) and the exact number depends on the particular plant from which the starch comes. On hydrolysis, glucose is the only monosaccharide produced.

Starch is often used as a starting point in fermentation. Clearly it has to be broken down into glucose units before it can be converted to ethanol. This has traditionally involved a fairly complex and costly industrial process, but more recent advancements have seen the use of bioconversion techniques. These incorporate an initial step called saccharification where starch is converted to glucose by the action of microorganisms or enzymes like glucoamylase and  $\alpha$ -amylase. This is followed by fermentation, where sugar is converted into ethanol using baker's yeast, *Saccharomyces cerevisiae*. Beer, whisky, and vodka can all be made from the fermentation of grain starches converted to sugar by the enzyme  $\alpha$ -amylase which is present in malted kernels.

Glycogen - is the only starch found in humans. It consists of several glucose units and is deposited in the liver as a reserve store of carbohydrate and in muscle as a more immediate source of fuel for energy.

Cellulose - is the chief structural carbohydrate of plants. With other materials, it provides the rigid and fibrous structure of fruits, vegetables and cereal grains, including the cell walls of the starch granules which unlike starch cannot be digested by humans and so are unavailable to the body and have no nutritional value. These indigestible polysaccharides are collectively known as dietary fibre or roughage and greatly aid the passage of food along the digestive tract and as bulk to the faeces.

Pectin - is another complex carbohydrate found mainly in fruits. It is responsible for the formation of gels in jam and jelly making.

## MICROBIOLOGY

On a commercial level, breweries and wineries are complex environments. Most large operations employ microbiologists with sound knowledge of fermentation processes and the biochemistry involved. Microbiology as applied to brewing incorporates knowing how microbes behave and how metabolites influence the properties of beverages. A good grounding in food science is also beneficial. Another aspect that is more practical is having good sensory skills for detecting problems with taste and flavour. Many microbiologists develop a sense for smelling different types of yeast. Microbiologists must be able to determine when a microorganism poses a threat to brewing.

## Yeasts

Yeasts are microscopic organisms that facilitate the fermentation process. They occur everywhere in nature, but there are many different types, ranging from wine yeasts to brewers' and bakers' yeasts. All have the ability to use sugar and create energy. Yeasts do not necessarily always produce alcohol, though. If yeast acts upon carbohydrates in the presence of ample oxygen, it tends to produce carbon dioxide, water, and heat. However, if oxygen is in limited or no supply, the action will produce alcohol.

There are numerous yeasts, but only a few are of great value in winemaking. For any yeast, it is important to consider acidity and growth. Wine yeasts must grow well in acidic solutions. Of all yeasts, *Saccharomyces cerevisiae* is the most important yeast in winemaking. It is preferred because it can tolerate high levels of sulphur dioxide and alcohol, thrives in a typical wine pH of 2.8 to 4, and it is vigorous when used in fermentation.

*S. cerevisiae* is not the only yeast involved in winemaking. In most cases, grapes also contain wild yeasts and it is these yeasts which often commence the fermentation process when the grapes are crushed. Wild yeasts include many different strains such as *Candida*, *Brettanomyces*, and *Kloeckera*. Generally, the action of these yeasts begins to diminish once sulphur dioxide is added and they tend to die off when the alcohol level of the fermenting must hit about 5%. This is when *S. cerevisiae* becomes more active.

## ENZYMES

Enzymes are chemical substances that can change other substances by a process called fermentation. Fermentation is the process which occurs in the brewing of beer and in the decomposition of dead matter. Enzymes are chemicals very like proteins in composition and they have three major properties:

- They can bring about chemical change in other substances without themselves being changed in the process.
- They can also achieve those changes inside the body in conditions of mild heat and comparatively mild acidity or alkalinity. In a laboratory, such changes would require conditions of great heat and the use of strong acids or alkalis.
- It takes only a very small amount of an enzyme to achieve big changes.

Enzymes are, therefore, very powerful chemicals indeed and they are specific to certain tasks. One particular enzyme will carry out one job and no other. Some enzymes can break down only some carbohydrates while others only act on protein.

Chemical reactions that would normally occur slowly outside of living tissue will often proceed much faster inside a living cell. Enzymes are a group of proteins that are involved in causing this "speeding up" of chemical reactions within cell tissue.

- Enzymes are proteins that are synthesised within a cell.
- Enzymes are catalysts to chemical reactions.
- Enzymes speed up chemical reactions so that the rate is more relevant (or compatible) with the cell and its needs.
- Enzymes work at extremely small concentrations, very much out of proportion to the chemical changes it influences.
- Enzymes are highly specific: every different reaction needs a quite different enzyme to be a catalyst. (There can be thousands of different enzymes, each influencing a different reaction).
- Enzymes lose their ability to catalyse reactions if subjected to certain changes including heat, extremes of pH, organic solvents or other chemicals that can denature protein.

The rate of an enzyme-affected reaction depends directly upon the quantity of enzyme present (in relation to the chemicals involved in the reaction); i.e., if more of the enzyme is added, the reaction is faster.

## Effect of Temperature

Enzyme-catalysed reactions are sensitive to changes in temperature. As temperature increases, there is an increase in the rate of reaction up to the level of perhaps 45-55°C. As the temperature rises, enzyme protein is damaged and, in turn, the reaction rate decreases.

## Effect of pH

If pH goes too high or too low, enzymes simply cease being effective. pH changes can have a significant effect upon the ionic nature of amino and carboxylic acid groups on the surface of proteins; hence significantly affecting the catalytic properties of enzymes. There are optimum pH levels at which certain enzyme catalysed chemical reactions will occur. The reaction mixture needs to be carefully controlled and buffered for reactions to occur normally.

Ideally, grape juice should contain about 0.6 to 1% of acid when it is harvested (i.e. 6 to 10 gm of acid per litre of grape juice). That level produces a wine that is satisfactory to most tastes. If acid levels are higher the wine produced is generally too acid for most people's taste.

The optimum pH for most yeasts to develop is 3.1 to 3.5. If pH drops too low, it can be raised by adding appropriate amounts of food grade calcium carbonate (or appropriate products containing calcium carbonate). If the pH is too high, it can be lowered by adding organic acids such as citric acid or tartaric acid. Small quantities of malic acid are also sometimes added to wines at the very end of fermentation to give the wine a slightly fruitier taste.

While wine may be made without adding enzymes; modern winemaking does frequently involve adding enzymes. They are naturally occurring chemicals produced by living things. Enzymes that are used in fermenting drinks are not created synthetically. They are extracted or harvested from living organisms. For example, *Aspergillus niger* is a fungus that grows commonly and naturally on fermentation containers like wine barrels. This fungus has been used in wine fermentation since at least the 19th century.

Enzymes are used by winemakers to reduce a biochemical affect that might otherwise cause a problem in the process of winemaking e.g. commercial enzymes are made up of enzymes from the pectinase family and are used to breakdown pectins (hemicellulose, cellulose).

Pectins are long-chain carbohydrates that have the potential to create a jelly-like component in your brewed drink, but by adding a "Pectinase" enzyme, these long-chain chemicals will be broken up, thus reducing the chance of a gel-forming. Glucans are another long-chained molecule that tends to occur as *Botrytis* mould develops on very ripe grapes. Adding an appropriate enzyme can cause a glucinase action that breaks up these glucan molecules.

Glycosides are smaller compounds that contain a component called a terpene. The glycoside typically becomes attached to glucose molecules in a fermented drink. The terpene component is important to giving wine an aroma. When the glucose and glycoside are bound together, the scent or aroma of a wine will diminish; but if that bond is broken, the terpene can become more active and the wine will have a stronger aroma. Adding the right enzyme to a wine can greatly improve its aroma.

Enzymes can be obtained from commercial suppliers, but it is important to ensure you are getting a high-quality product.

## QUALITY CONTROL

### Containers

Stainless steel is the ideal choice for containers. Concrete and fibreglass which has been specially coated is also fine. Oak casks are traditional winemaking vessels, but they must be treated before they can be used, and they tend to be very expensive. Copper and iron are never used.

Containers must never be filled to the top but instead about one third of the depth is left free above the must to allow for fermentation. During this process, grape skins will float to the surface and there must be space to allow for the mixture to be stirred several times per day.

Containers must be sealed to keep out oxygen. This can be achieved using a pressure valve in the form of a cork or rubber bung with a plastic tube attached to a vessel containing water. Water immersed in the tube is a fermentation trap. It allows carbon dioxide to escape and prevents air from getting in. When fermentation is complete, there is a drop in the release rate of CO<sub>2</sub> until

very quickly no more is released.

If a must get stuck and fermentation is incomplete it must be restarted as soon as possible. Sometimes this can be resolved by adding fermenting must from another batch. Sometimes aeration can be of help.

## Sweetness

Almost all wine yeasts become inactive when alcohol levels reach 14 to 15%. At this point, fermentation ceases. The amount of sugar that remains in the wine has a direct bearing on its sweetness.

| Remaining Sugar (%) | Type of Wine |
|---------------------|--------------|
| > 4                 | Sweet        |
| 2                   | Semi-sweet   |
| < 1                 | Dry          |

Kits can be used to test the amount of residual sugar left in a wine. A wine can be sweetened after fermentation by adding unfermented grape juice. To be successful, the yeast has to be deactivated before the grape juice is added; otherwise, fermentation will recommence. This is achieved by sterile filtration.

## Malolactic Fermentation

As outlined earlier, some wines are put through a secondary fermentation called malolactic fermentation, often abbreviated to MLF or simply 'malo' amongst winemakers. It is used widely in the production of dry wine, particularly full-bodied whites and medium to full-bodied reds. It is also used in many of the cooler wine making regions since wines from these regions often have naturally higher levels of malic acid. This is because the growth of bacteria responsible for this secondary fermentation is inhibited by low temperatures and the presence of sulphur dioxide. In such cases, malolactic fermentation becomes a necessity. As the name implies, malolactic fermentation results in the conversion of malic acid into lactic acid. This is achieved by lactic acid bacteria (LAB), particularly *Oenococcus oeni*, and some other species of *Pediococcus* and *Lactobacillus*.

Malolactic fermentation is desirable because it reduces the tart taste associated with malic acid. If malic acid is present in large enough quantities, then a wine will taste sour or too acidic. Although lactic acid is also acidic, it is preferable because it has a more agreeable and smoother taste. Lactic acid also has a softer mouth feel compared to the hardness or metallic taste of malic acid.

Another outcome of malolactic fermentation is the release of small amounts of carbon dioxide, ethanol, and acetic acid. Winemakers aim to have this happen before the wine is bottled because even though the amount of carbon dioxide emitted is nominal it can cause small bubbles to appear and will leave a fizzy taste in the mouth if consumed. If malolactic fermentation has already taken place in the barrel it is much less likely to occur in bottles, which is another good reason to have wines undergo secondary fermentation. Sometimes wine undergoes malolactic fermentation in the bottle and has an obvious fizz when poured. Wine like this appears to be still fermenting and can taste quite awful.

Malolactic fermentation occurs due to certain types of bacteria in the wine. These bacteria fall into three genera:

- *Oenococcus* (formerly *Leuconostoc*)
- *Pediococcus*
- *Lactobacillus*

Bacteria in these genera can be found in most wineries. They are often introduced through handling. The activity of these LABs often improves the end wine's flavour profile by making it smoother and more complex so that it is well-balanced and more drinkable. However, if some of these LABs are too dominant, they can make the wine taste bad. For example, some *Lactobacillus* species are reputed to impart sweaty flavours or those associated with sour milk. Others are known to cause what is commonly called 'mousy taint', regarded as the worst fault a wine can have. Usually, though, it is *Oenococcus oeni*, which is most dominant because it can tolerate higher levels of alcohol and acidity than the others, and it produces substances with much more agreeable flavours and aromas. One of these substances is diacetyl, which gives the wine 'butteryness'. Nevertheless, even this can cause problems if too much is present because the soft butter taste becomes more like sour butter.

## Controlling Factors

Not all wines benefit from malolactic fermentation. If you desire a crisp, acidic wine such as a Riesling, then MLF would ruin the taste. To prevent MLF, winemakers can keep temperatures cool during winemaking and bottling and control levels of the antimicrobial agent sulphur dioxide, which inhibits LABs. Winemakers wishing to start MLF can speed up the process using starter



cultures. This is achieved through inoculating the wine with some from a cask that is already undergoing MLF or by adding a readymade culture.

Other ways to induce MLF include:

- Temperature - malolactic fermentation will be inhibited at temperatures below 15°C because LABs will fail to grow and become inactive. On the other hand, if the temperature is allowed to reach 40°C or more then whilst the bacteria may grow well the wine's character can be harmed.
- pH levels - a PH range of 3.2. to 3.4 must be maintained, or adjusted to fall within this range. Outside this range LABs will not be active.
- Nutrients - the LABs need something to feed on. For this reason, the wine is not filtered until after malolactic fermentation, so the sediment or lees acts as a food source. However, the wine should not be left unfiltered for too long to avoid introducing off flavours.
- Oxygen - some aeration is needed to encourage the growth of good bacteria, but this should be minimal. Wine may be racked just once to create the desired level of aeration.
- Sulphur dioxide - levels of SO<sub>2</sub> should not be introduced as it will restrict growth of LABs. Similarly, carbon dioxide and nitrogen should not be introduced.

## SECONDARY (MALO LACTIC) FERMENTATION

This is a process that converts malic acid into lactic acid, and at the same time creates small quantities of acetic acid, ethanol and carbon dioxide. The process results in an overall reduction in acidity of wine and can give the wine greater stability extending the time for bottle aging. The process is driven by the action of several species of bacteria which commonly exist in wineries (ie. Genera *Leuconostoc*, *Pediococcus* and *Lactobacillus*)

This is not an easy process to control, and winemakers should be very familiar with other winemaking operations before attempting to use secondary fermentation. It is most commonly used with red wines in cool climates.

To facilitate malo lactic fermentation the following conditions are needed:

- pH of 3.2 to 3.4
- Temperature needs to be over 15°C so that bacterial growth is not slowed.
- Sediment needs to be present. After primary fermentation, filtering needs to be delayed and sediment left to retain nutrients needed for a secondary fermentation
- Slight aeration is needed to encourage bacteria
- Avoid/delay adding sulphur dioxide as it can kill the bacteria
- Wine that is already undergoing malolactic fermentation (thus has appropriate bacteria) can be added to start this secondary fermentation. (5% already fermenting to 95% starting the secondary process).